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Avelios Medical GmbH
Munich, Germany

Re: Working Student Machine Learning (all genders)

Dear Hiring Team,

I'm applying for the Working Student Machine Learning role at Avelios. Putting language models and retrieval into the daily workflow of clinicians, where a wrong answer has a real cost, is the problem I want to work on. I'm an MSc Data Science student in Germany with hands-on LLM, RAG and embedding projects, and I can start as a working student now.

The closest match to your "RAG, embeddings and evaluation" work is my RAG pipeline. I built the retriever by hand through six gated stages (hybrid BM25 and dense search on pgvector with BGE-M3), then rebuilt it in LangChain and benchmarked both on a 55-question golden set. That comparison caught a silent 0.06 drop in recall caused by a default tokenizer, which I would not have found without the benchmark. I then added a corrective loop that re-retrieves on a bad match and refuses when the corpus cannot answer. It reaches 100% correct refusal, which matters more in a hospital than a confident wrong answer.

On the training side, I built and trained a CLIP ViT-B/32 and DistilBERT dual-encoder in PyTorch for my MSc project. It uses a symmetric InfoNCE loss into a shared 1024-dimensional space and reaches a validation median rank of 10 out of 200 on a 13.5K-image dataset. Caching the CLIP image features cut training time by roughly 10x, so I could iterate on the text side faster. For reliability, JobPilot, my 16-node LangGraph pipeline, blocks any output that lacks evidence in a Chroma-backed store, and 238 pytest tests keep it honest. Before the MSc I spent 2.5 years as a software engineer on regulated insurance data, where I resolved 100+ production defects, so shipping and debugging live systems is familiar ground.

Some gaps, stated plainly. I have not worked with speech recognition, and I have not built federated learning systems. Privacy-preserving training on hospital data is one of the reasons your posting interests me, and I would like to learn it on real problems. My ML work so far is from my MSc and personal projects, not commercial production, and my German is A1/A2 while I keep learning it. Python, PyTorch, Hugging Face transformers and RAG with vector databases are where I can contribute from the first week.

I'm available as a working student now, open to relocating to Munich, and would like to grow into a full-time role after my MSc. I'd welcome the chance to talk through how I could help Avelios bring models into production.

Thank you for your time and consideration.

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